

Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-1180
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	ServiceNow Service Portal (Employee Center) with Service Catalog forms; HTML/CSS/AngularJS widgets	Employees order laptops from a ready-made, responsive portal with no separate front-end build. Catalog variables and UI policies give a simple, guided order form.
2	Backend / Server-Side	ServiceNow platform: Flow Designer, Workflow Studio and server-side JavaScript (Business Rules, Script Includes)	Flow Designer automates validation, approval, PO creation and notification with low-code logic. Server-side JavaScript handles the few custom rules, so no separate backend is needed.
3	Database / Data Storage	ServiceNow Now Platform database (Request, Requested Item, Approval, Purchase Order and CMDB asset tables)	Built-in relational store with ready tables for requests, approvals and assets. Keeps all procurement data and audit history in one place.
4	Cloud / Hosting / Deployment	ServiceNow cloud instance (Personal Developer Instance for development and testing)	Fully managed SaaS hosting with built-in scalability, backups and high availability. Free PDI suits development; Update Sets move changes between instances.
5	Version Control & CI/CD	ServiceNow Update Sets with Source Control integration (GitHub); ATF (Automated Test Framework)	Tracks and moves configuration changes safely and supports team collaboration. ATF automatically tests the catalog item and flow after each change.

6	Third-Party APIs / Other Tools	ServiceNow REST API / Integration Hub (Vendor ordering), Email and Microsoft Teams notifications, Azure AD / SSO	Sends purchase orders to the vendor and receives confirmations. Notifies employees and managers at each stage. SSO gives secure employee login and manager lookup.
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